

ALGEBRA

Quadratic Equation

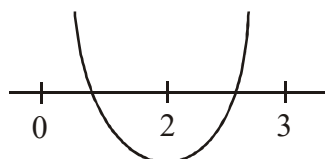
1. Consider the quadratic equation $(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval (0, 2) and its other root lies in the interval (2, 3). Then the number of elements in S is :

द्विघातीय समीकरण $(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$ पर विचार कीजिए। माना S, c के उन सभी पूर्णांकीय मानों, जिनके लिए समीकरण का एक मूल अंतराल (0, 2) में है तथा इसका दूसरा मूल अंतराल (2, 3) में है, का समुच्चय है, तो S के अवयवों की संख्या है :

- (A) 11 (B) 10 (C) 12 (D) 18

Ans. A

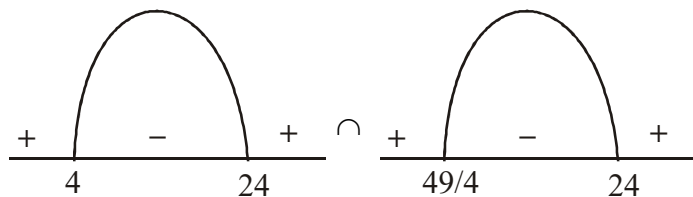
Sol.



$$f(x) = (c - 5)x^2 - 2cx + c - 4$$

$$f(0) \cdot f(2) < 0 \dots\dots (1) \quad \& \quad f(2) \cdot f(3) < 0$$

$$(c - 4)(c - 24) < 0 \quad \& \quad (c - 24)(4c - 49) < 0$$



$$c \in \left(\frac{49}{4}, 24 \right) = 13, 14, \dots\dots 23$$

No of I = 11

2. If λ be the ratio of the roots of the quadratic equation in x, $3m^2x^2 + m(m - 4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$ is :

- (A) $2 - \sqrt{3}$ (B) $-2 + \sqrt{2}$ (C) $4 - 3\sqrt{2}$ (D) $4 - 2\sqrt{3}$

यदि x में द्विघात समीकरण $3m^2x^2 + m(m - 4)x + 2 = 0$ के मूलों का अनुपात λ है, तो m का वह न्यूनतम मान जिसके लिए

$$\lambda + \frac{1}{\lambda} = 1 \text{ है, है -}$$

- (A) $2 - \sqrt{3}$ (B) $-2 + \sqrt{2}$ (C) $4 - 3\sqrt{2}$ (D) $4 - 2\sqrt{3}$

Ans. C

Sol. $3m^2x^2 + m(m - 4)x + 2 = 0$

$$\text{Ratio of roots} \Rightarrow \frac{\alpha}{\beta} = \lambda$$

$$\begin{aligned}\frac{\alpha}{\beta} + \frac{\beta}{\alpha} &= \frac{\alpha^2 + \beta^2}{\alpha\beta} = 1 \\ &= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = 1 \\ &= \left(\frac{m(m-4)}{3m^2} \right)^2 - \frac{4}{3m^2} = \frac{2}{3m^2} \\ &= m^2(m-4)^2 - 6.3m^2 = 0 \\ &= (m-4)^2 - 18 = 0 \\ m &= 4 - 3\sqrt{2}\end{aligned}$$

3. If m is chosen in the quadratic equation $(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$ such that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is:

(A) $8\sqrt{3}$ (B) $10\sqrt{5}$ (C) $4\sqrt{3}$ (D) $8\sqrt{5}$

यदि द्विघातीय समीकरण $(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$ में m इस प्रकार लिया गया है, कि इसके मूलों का योगफल अधिकतम है, तो इसके मूलों के घन का निरपेक्ष अन्तर है :

(A) $8\sqrt{3}$ (B) $10\sqrt{5}$ (C) $4\sqrt{3}$ (D) $8\sqrt{5}$

Ans. D

Sol. Sum of roots = $\frac{3}{m^2 + 1}$

For maximum $m = 0$

Hence equation becomes $x^2 - 3x + 1 = 0$

$\alpha + \beta = 3, \alpha\beta = 1, |\alpha - \beta| = \sqrt{5}$

$|\alpha^3 - \beta^3| = (\alpha - \beta)(\alpha^2 + \beta^2 + \alpha\beta) = \sqrt{5}(9 - 1) = 8\sqrt{5}$

4. If α and β are the roots of the quadratic equation, $x^2 + x \sin\theta - 2\sin\theta = 0$, $\theta \in \left(0, \frac{\pi}{2}\right)$, then

$\frac{\alpha^{12} + \beta^{12}}{(\alpha^{-12} + \beta^{-12}) \cdot (\alpha - \beta)^{24}}$ is equal to :

(A) $\frac{2^{12}}{(\sin\theta - 8)^6}$ (B) $\frac{2^{12}}{(\sin\theta - 4)^{12}}$ (C) $\frac{2^6}{(\sin\theta + 8)^{12}}$ (D) $\frac{2^{12}}{(\sin\theta + 8)^{12}}$

यदि द्विघात समीकरण, $x^2 + x \sin\theta - 2\sin\theta = 0$, $\theta \in \left(0, \frac{\pi}{2}\right)$ के मूल α तथा β हैं, तो

$\frac{\alpha^{12} + \beta^{12}}{(\alpha^{-12} + \beta^{-12}) \cdot (\alpha - \beta)^{24}}$ बराबर है :

(A) $\frac{2^{12}}{(\sin \theta - 8)^6}$ (B) $\frac{2^{12}}{(\sin \theta - 4)^{12}}$ (C) $\frac{2^6}{(\sin \theta + 8)^{12}}$ (D) $\frac{2^{12}}{(\sin \theta + 8)^{12}}$

Ans. D

Sol. Given $\alpha + \beta = -\sin \theta$ and $\alpha\beta = -2\sin \theta$

$$\frac{(\alpha^{12} + \beta^{12})\alpha^{12}\beta^{12}}{(\alpha^{12} + \beta^{12})(\alpha - \beta)^{24}} = \frac{(\alpha\beta)^{12}}{(\alpha - \beta)^{24}}$$

$$|\alpha - \beta| = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \sqrt{\sin^2 \theta + 8\sin \theta}$$

Hence required quantity

$$\frac{(\alpha\beta)^{12}}{(\alpha - \beta)^{24}} = \frac{(2\sin \theta)^{12}}{\sin^{12} \theta (\sin \theta + 8)^{12}} = \frac{2^{12}}{(\sin \theta + 8)^{12}}$$

5. If α , β and γ are three consecutive terms of a non constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root then $\alpha(\beta + \gamma)$ is equal to

(A) 0 (B) $\alpha\gamma$ (C) $\beta\gamma$ (D) $\alpha\beta$

यदि एक भिन्न पदों वाली गुणोत्तर श्रेणी के तीन क्रमागत पद α , β तथा γ इस प्रकार हैं कि समीकरणों $\alpha x^2 + 2\beta x + \gamma = 0$ तथा $x^2 + x - 1 = 0$ का एक मूल समान है, तो $\alpha(\beta + \gamma)$ बराबर है :

(A) 0 (B) $\alpha\gamma$ (C) $\beta\gamma$ (D) $\alpha\beta$

Ans. C

Sol. $\beta^2 = \alpha\gamma$ so roots of the equation $\alpha x^2 + 2\beta x + \gamma = 0$

$$\text{are } \frac{-2\beta \pm 2\sqrt{\beta^2 - \alpha\gamma}}{2\alpha} = -\frac{\beta}{\alpha}$$

This root satisfy the equation $x^2 + x - 1 = 0$

$$\beta^2 - \alpha\beta - \alpha^2 = 0$$

$$\Rightarrow \alpha\gamma - \alpha\beta - \alpha^2 = 0$$

$$\Rightarrow \alpha + \beta = \gamma$$

$$\begin{aligned} \text{Now, } \alpha(\beta + \gamma) &= \alpha\beta + \alpha\gamma = \alpha\beta + \beta^2 \\ &= (\alpha + \beta)\beta = \beta\gamma \end{aligned}$$

6. Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true ?

(A) $p_5 = p_2 \cdot p_3$ (B) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$

(C) $p_3 = p_5 - p_4$ (D) $p_5 = 11$

माना α तथा β समीकरण $x^2 - x - 1 = 0$ के मूल हैं। यदि $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, तो निम्न में से कौनसा एक कथन सत्य नहीं है?

(A) $p_5 = p_2 \cdot p_3$

(B) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$

(C) $p_3 = p_5 - p_4$

(D) $p_5 = 11$

Ans. A

Sol. $x^2 - x - 1 = 0$ roots are λ, β .

$$P_k = \alpha^k + \beta^k$$

$$= \alpha^{k-2}(\alpha + 1) + \beta^{k-2}(\beta + 1)$$

$$= (\alpha^{k-1} + \beta^{k-1}) + (\alpha^{k-2} + \beta^{k-2})$$

$$p_k = p_{k-1} + p_{k-2} \quad \dots\dots\dots(1)$$

$$p_1 = \alpha + \beta = 1, \quad p_2 = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 3$$

$$p_3 = p_2 + p_1 = 4$$

$$p_4 = p_3 + p_2 = 7$$

$$p_5 = p_4 + p_3 = 11$$

Now check option

7. Let $a, b \in \mathbf{R}, a \neq 0$ be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α , which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to :

(A) 28

(B) 25

(C) 26

(D) 24

माना $a, b \in \mathbf{R}, a \neq 0$ इस प्रकार हैं कि समीकरण $ax^2 - 2bx + 5 = 0$ का α पुनरावृत्त मूल है, जो समीकरण $x^2 - 2bx - 10 = 0$ का भी एक मूल है। यदि β इस समीकरण का दूसरा मूल है, तो $\alpha^2 + \beta^2$ बराबर है :

(A) 28

(B) 25

(C) 26

(D) 24

Ans. B

Sol. $ax^2 - 2bx + 5 = 0$ $\begin{matrix} \nearrow \alpha \\ \searrow \alpha \end{matrix}$

$$\alpha = \frac{b}{a}$$

$$\alpha^2 = \frac{5}{a}$$

$$b^2 = 5a$$

$$x^2 - 2bx - 10 = 0 \quad \begin{matrix} \nearrow \alpha \\ \searrow \beta \end{matrix}$$

$$\alpha + \beta = 2b \quad \alpha\beta = -10$$

$$\frac{b^2}{a^2} - \frac{2b^2}{a} - 10 = 0$$

$$b^2 - 2ab^2 - 10a^2 = 0$$

$$a = \frac{1}{4}$$

$$b^2 = \frac{5}{4}$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 4b^2 + 20 = 4 \times \frac{5}{4} + 20 = 25$$

8. Let $f(x)$ be a quadratic polynomial such that $f(-1) + f(2) = 0$. If one of the roots of $f(x) = 0$ is 3, then its other root lies in :

माना $f(x)$ एक द्विघात बहुपद है जिसके लिए $f(-1) + f(2) = 0$ है। यदि $f(x) = 0$ का एक मूल 3 है, तो दूसरा मूल निम्न में से किस अन्तराल में स्थित है?

- (A) $(-3, -1)$ (B) $(0, 1)$ (C) $(1, 3)$ (D) $(-1, 0)$

Ans. D

Sol. $f(3) = 0$

$$\Rightarrow f(x) = k(x-3)(x-\alpha)$$

$$f(2) + f(-1) = 0$$

$$k(-1)(-2-\alpha) + k(-4)(-1-\alpha) = 0$$

$$k(5\alpha + 2) = 0$$

$$\therefore k \neq 0$$

$$\Rightarrow \alpha = \frac{-2}{5}$$

$$\frac{-2}{5} \in (-1, 0).$$

9. If α and β are the roots of equation $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation

$$2x^2 + 2qx + 1 = 0, \text{ then } \left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right) \text{ is equal to :}$$

यदि α तथा β , समीकरण $x^2 + px + 2 = 0$ के मूल हैं तथा $\frac{1}{\alpha}$ तथा $\frac{1}{\beta}$, समीकरण $2x^2 + 2qx + 1 = 0$ के मूल हैं, तो

$$\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right) \text{ बराबर है :}$$

- (A) $\frac{9}{4}(9-q^2)$ (B) $\frac{9}{4}(9+p^2)$ (C) $\frac{9}{4}(9-p^2)$ (D) $\frac{9}{4}(9+q^2)$

Ans. C

Sol. $x^2 + px + 2 = 0$ $\begin{matrix} \nearrow \alpha \\ \searrow \beta \end{matrix}$

$$\alpha + \beta = -p$$

$$\alpha \cdot \beta = 2$$

$$\begin{aligned} & \left(\alpha + \frac{1}{\beta} \right) \left(\beta + \frac{1}{\alpha} \right) \left(\alpha - \frac{1}{\alpha} \right) \left(\beta - \frac{1}{\beta} \right) \\ &= \left(\alpha\beta + \frac{1}{\alpha\beta} + 2 \right) \left(\alpha\beta + \frac{1}{\alpha\beta} - \left(\frac{\beta}{\alpha} + \frac{\alpha}{\beta} \right) \right) \\ &= \left(2 + \frac{1}{2} + 2 \right) \left(2 + \frac{1}{2} - \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} \right) \\ &= \frac{9}{2} \left(\frac{5}{2} - \frac{p^2 - 4}{2} \right) = \frac{9}{4} (9 - p^2) \end{aligned}$$

10. If α and β are the roots of the equation $2x(2x + 1) = 1$, then β is equal to :

यदि α तथा β समीकरण $2x(2x + 1) = 1$ के मूल हैं, तो β बराबर है :

- (A) $2\alpha(\alpha - 1)$ (B) $2\alpha^2$ (C) $2\alpha(\alpha + 1)$ (D) $-2\alpha(\alpha + 1)$

Ans. D

Sol. $2x(2x + 1) = 1$

$$4x^2 + 2x - 1 = 0$$

$$4\alpha^2 + 2\alpha - 1 = 0 \Rightarrow 4\alpha^2 + 2\alpha = 1$$

$$2\alpha^2 + \alpha = \frac{1}{2}$$

$$\alpha + \beta = -\frac{1}{2}$$

$$\beta = -\alpha - \frac{1}{2}$$

$$\beta = -\alpha - (2\alpha^2 + \alpha)$$

$$\beta = -2\alpha - 2\alpha^2 = -2\alpha(\alpha + 1)$$

11. The integer 'k', for which the inequality $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$ is valid for every x in $\mathbb{R}$, is :

- (A) 0 (B) 4 (C) 2 (D) 3

पूर्णांक 'k', जिसके लिए असमिका $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$, R में प्रत्येक x के लिए मान्य है, है :

Ans. D

Sol. $D < 0$

$$4(3k - 1)^2 - 4(8k^2 - 7) < 0$$

$$9k^2 - 6k + 1 - 8k^2 + 7 < 0$$

$$k^2 - 6k + 8 < 0$$

$$(k - 4)(k - 2) < 0$$

$$k \in (2, 4)$$

$$\Rightarrow k = 3$$

12. The value of $3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$ is equal to :

$3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$ का मान बराबर है :

(A) $3 + 2\sqrt{3}$

(B) $2 + \sqrt{3}$

(C) $4 + \sqrt{3}$

(D) $1.5 + \sqrt{3}$

Ans. D

Sol. Let $x = 3 + \frac{1}{4 + \frac{1}{x}}$

$$\Rightarrow x = 3 + \frac{x}{4x + 1}$$

$$\Rightarrow x = \frac{13x + 3}{4x + 1}$$

$$\Rightarrow 4x^2 - 12x - 3 = 0$$

$$\Rightarrow x = \frac{12 \pm \sqrt{144 + 48}}{8} = \frac{3 \pm 2\sqrt{3}}{2}$$

Since $x > 0$

so $x = 1.5 + \sqrt{3}$

13. The number of real solutions of the equation $3\left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) + 5 = 0$, is :

समीकरण $3\left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) + 5 = 0$, के वास्तविक हलों की संख्या है :

- (A) 4 (B) 0 (C) 2 (D) 3

Ans. B

Sol. $3\left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) + 5 = 0$
 $3\left[\left(x + \frac{1}{x}\right)^2 - 2\right] - 2\left(x + \frac{1}{x}\right) + 5 = 0$

Let $x + \frac{1}{x} = t$; $t \in (-\infty - 2] \cup [2, \infty)$

$3t^2 - 6 - 2t + 5 = 0$

$3t^2 - 2t - 1 = 0$

$t = 1, t = \frac{-1}{3}$ But $t \in (-\infty, -2] \cup [2, \infty)$

$\Rightarrow$ No. real solution

14. Let $\lambda \neq 0$ be a real number. Let α, β be the roots of the equation $14x^2 - 31x + 3\lambda = 0$ and α, γ be the roots of the equation $35x^2 - 53x + 4\lambda = 0$. Then $\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation

माना $\lambda \neq 0$ एक वास्तविक संख्या है। माना α, β समीकरण $14x^2 - 31x + 3\lambda = 0$ के मूल हैं एवं α, γ समीकरण

$35x^2 - 53x + 4\lambda = 0$ के मूल है। तब $\frac{3\alpha}{\beta}$ व $\frac{4\alpha}{\gamma}$ किस समीकरण के मूल है ?

- (A) $49x^2 - 245x + 250 = 0$ (B) $7x^2 + 245x - 250 = 0$
 (C) $49x^2 + 245x + 250 = 0$ (D) $7x^2 - 245x + 250 = 0$

Ans. A

Sol. $14\alpha^2 - 31\alpha + 3\lambda = 0$... (i)
 $35\alpha^2 - 53\alpha + 4\lambda = 0$... (ii)
 $\therefore \alpha$ will satisfy both give equations

$$\therefore 14\alpha^2 - 31\alpha + 3\lambda = 0 \Rightarrow \alpha^2 - \frac{31\alpha}{14} + \frac{3\lambda}{14} = 0 \quad \dots (iii)$$

$$35\alpha^2 - 53\alpha + 4\lambda = 0 \Rightarrow \alpha^2 - \frac{53\alpha}{35} + \frac{4\lambda}{35} = 0 \quad \dots (iv)$$

from (iii) – (iv) we get

$$\lambda = 7\alpha \text{ put in } \dots \quad (iii)$$

$$\alpha^2 - \frac{31\alpha}{14} + \frac{3}{14}(7\alpha) = 0$$

$$\alpha^2 - \frac{31\alpha}{14} + \frac{3\alpha}{2} = 0$$

$$\Rightarrow \alpha = 0 \text{ or } \alpha = \frac{5}{7}$$

($\alpha = 0$ not acceptable as $\lambda \neq 0$)

$$\therefore \alpha = \frac{5}{7}$$

$$\therefore \alpha + \beta = \frac{31}{14} \Rightarrow \beta = \frac{3}{2}$$

$$\therefore \alpha + \gamma = \frac{53}{35} \Rightarrow \gamma = \frac{4}{5}$$

$$\therefore \frac{3\alpha}{\beta} = \frac{10}{7} \text{ and } \frac{4\alpha}{\gamma} = \frac{25}{7}$$

$$\therefore \text{Reqd. Q.E. is } \left(x - \frac{10}{7}\right)\left(x - \frac{25}{7}\right) = 0$$

$$(7x - 10)(7x - 25) = 0$$

$$49x^2 - 245x + 250 = 0$$

15. The number of real roots of the equation $\sqrt{x^2 - 4x + 3} + \sqrt{x^2 - 9} = \sqrt{4x^2 - 14x + 6}$, is :

समीकरण $\sqrt{x^2 - 4x + 3} + \sqrt{x^2 - 9} = \sqrt{4x^2 - 14x + 6}$ के वास्तविक मूलों की संख्या है :

(A) 3

(B) 0

(C) 1

(D) 2

Ans. C

Sol. $\sqrt{(x-1)(x-3)} + \sqrt{(x-3)(x+3)} = \sqrt{2(2x-1)(x-3)}$

$$x - 3 = 0 \text{ and } \sqrt{x-1} + \sqrt{x+3} = \sqrt{2(2x-1)} \quad \dots (1)$$

$$\text{Squaring both side; } x - 1 + x + 3 + 2\sqrt{(x-1)(x+3)} = 4x - 2$$

$$2\sqrt{(x-1)(x+3)} = 2x - 4$$

$$\sqrt{x^2 + 3x - x - 3} = x - 2 \quad \dots\dots (2)$$

Squaring both side; $x^2 + 2x - 3 = x^2 - 4x + 4$

$$6x = 7 \Rightarrow x = \frac{7}{6}$$

At $x = \frac{7}{6}$ equation (2) positive = negative

Rejected

$\therefore$ real root $x = 3$

16. The equation $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^x + 1 = 0, x \in \mathbb{R}$ has :

- (A) four solutions two of which are negative
- (B) two solutions and only one of them is negative
- (C) two solutions and both are negative
- (D) no solution

समीकरण $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^x + 1 = 0, x \in \mathbb{R}$:

- (A) के चार हल हैं जिनमें से दो ऋणात्मक है
- (B) के दो हल हैं तथा उनमें से केवल एक ऋणात्मक है
- (C) के दो हल हैं जिनमें से दो ऋणात्मक है
- (D) का कोई हल नहीं है

Ans. C

Sol. $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^x + 1 = 0$

Let $e^x = t$

Now $t^4 + 8t^3 + 13t^2 - 8t + 1 = 0$

Dividing equation by t^2 ,

$$t^2 + 8t + 13 - \frac{8}{t} + \frac{1}{t^2} = 0$$

$$t^2 + \frac{1}{t^2} + 8\left(t - \frac{1}{t}\right) + 13 = 0$$

$$\left(t - \frac{1}{t}\right)^2 + 2 + 8\left(t - \frac{1}{t}\right) + 13 = 0$$

Let $t - \frac{1}{t} = z$

$$z^2 + 8z + 15 = 0$$

$$(z + 3)(z + 5) = 0$$

$$z = -3 \text{ or } z = -5$$

$$\text{So, } t - \frac{1}{t} = -3 \text{ or } t - \frac{1}{t} = -5$$

$$t^2 + 3t - 1 = 0 \text{ or } t^2 + 5t - 1 = 0$$

$$t = \frac{-3 \pm \sqrt{13}}{2} \text{ or } t = \frac{-5 \pm \sqrt{29}}{2}$$

as $t = e^x$ so t must be positive,

$$t = \frac{\sqrt{13} - 3}{2} \text{ or } \frac{\sqrt{29} - 5}{2}$$

$$\text{So, } x = \ln\left(\frac{\sqrt{13} - 3}{2}\right) \text{ or } x = \ln\left(\frac{\sqrt{29} - 5}{2}\right)$$

Hence two solution and both are negative.

17. The number of integral values of k , for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval $(1, 2)$ and its other roots lies in the interval $(2, 3)$, is :

k के पूर्णांक मानों, जिनके लिए समीकरण $2x^2 - 8x + k = 0$ का एक मूल अंतराल $(1, 2)$ में है, तथा दूसरा मूल अंतराल $(2, 3)$ में, की संख्या है :

- (A) 0 (B) 3 (C) 1 (D) 2

Ans. C

Sol. Let $f(x) = 2x^2 - 8x + k = 0$

$$f(1) > 0$$

$$f(2) < 0$$

$$f(3) > 0$$

$$f(1) > 0 \Rightarrow 2 - 8 + k > 0 \Rightarrow k > 6 \quad (A)$$

$$f(2) < 0 \Rightarrow 8 - 16 + k < 0 \Rightarrow k < 8 \quad (B)$$

$$f(3) > 0 \Rightarrow 18 - 24 + k > 0 \Rightarrow k > 6 \quad (C)$$

$$k \in (6, 8) \therefore \text{integral } k = 7$$

18. The sum of all the roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is :

समीकरण $|x^2 - 8x + 15| - 2x + 7 = 0$ के सभी मूलों का योग है :

- (A) $11 + \sqrt{3}$ (B) $11 - \sqrt{3}$ (C) $9 + \sqrt{3}$ (D) $9 - \sqrt{3}$

Ans. C

Sol. $|x^2 - 8x + 15| - 2x + 7 = 0$

$$|(x-3)(x-5)| - 2x + 7 = 0$$

Case – I : If $x \in (-\infty, 3] \cup [5, \infty)$

$$x^2 - 8x + 15 - 2x + 7 = 0$$

$$x^2 - 10x + 22 = 0$$

$$x = \frac{10 \pm \sqrt{12}}{2} = 5 \pm \sqrt{3}$$

$\therefore x = 5 + \sqrt{3}$ is accepted

Case – II : If $x \in (3, 5)$

$$-x^2 + 8x - 15 - 2x + 7 = 0$$

$$x^2 - 6x + 8 = 0$$

$x = 2, 4 \Rightarrow x = 4$ is only accepted

$\therefore$ Sum of roots $5 + \sqrt{3} + 4 = 9 + \sqrt{3}$

19. Let α, β, γ be the three roots of the equation $x^3 + bx + c = 0$. if $\beta\gamma = 1 = -\alpha$, then $b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$ is equal to:

माना α, β, γ समीकरण $x^3 + bx + c = 0$ के तीन मूल हैं। यदि $\beta\gamma = 1 = -\alpha$, तो $b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$ बराबर है।

- (A) 21 (B) 19 (C) $\frac{169}{8}$ (D) $\frac{155}{8}$

Ans. B

Sol. Roots of $x^3 + bx + c = 0$ are α, β, γ

$$\therefore \beta\gamma = 1 = -\alpha$$

$$\therefore \alpha = -1 \quad \dots(i)$$

$$\text{and } \alpha + \beta + \gamma = 0 \quad \dots(ii)$$

$$\alpha\beta\gamma = -c \quad \dots(iii)$$

$$\therefore c = 1 \quad \dots(iv)$$

$$\beta + \gamma = 1 \quad \dots(v)$$

$$\therefore \alpha\beta + \beta\gamma + \gamma\alpha = b$$

$$\Rightarrow \alpha(\beta + \gamma) + \beta\gamma = b$$

$$\therefore b = 0 \quad \dots(vi)$$

$$\therefore \beta = -\omega, \gamma = -\omega^2 \quad \dots(vii)$$

$$b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$$

$$= 0 + 2 + 3 + 6 + 8$$

$$= 19$$

20. Let α, β be the roots of the quadratic equation $x^2 + \sqrt{6}x + 3 = 0$. Then $\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$ is equal to :

माना द्विघातीय समीकरण $x^2 + \sqrt{6}x + 3 = 0$ के मूल α, β हैं। तो $\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$ बराबर है :

(A) 72

(B) 81

(C) 729

(D) 9

Ans. D

Sol.
$$\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$$

Let $a_n = \alpha^n + \beta^n$

$$= \frac{a_{23} + a_{14}}{a_{15} + a_{10}}$$

$$x^2 + \sqrt{6}x + 3 = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

$$x^2 + \sqrt{6}x + 3 = 0 \sum_{\beta}^{\alpha}$$

$$x = \frac{-\sqrt{6} \pm \sqrt{-6}}{2}$$

$$= \sqrt{6} \left(\frac{-1 \pm i}{2} \right)$$

$$= \sqrt{3} \left(\frac{-1 \pm i}{\sqrt{2}} \right)$$

$$\alpha = \sqrt{3} e^{\frac{i3\pi}{4}}$$